

BioMechAI: The 9 Official Modules

Complete Quick-Reference Defense Guide for FYP-II Final Presentation

Plain English Guide for Quick Answers to Evaluators and Panelists

Module 1: User Registration, Login & Role Security

ACCOUNT
SYSTEM

In Simple Words: Secure sign-up and sign-in system that separates Athletes from Coaches and verifies real emails.

HOW IT WORKS IN THE APP

- **Role Separation:** Athletes use the Flutter mobile application; coaches use the web portal.
- **Email Verification Gatekeeper:** When someone registers, Google sends a verification link. Unverified accounts cannot enter the app until they confirm their email.
- **Cloud Database:** Securely backed by Google Firebase Firestore, storing user settings and workout histories.

Presentation Answer Tip: If asked, explain that user security is hardened with mandatory email verification to block fake accounts, and athletes are dynamically routed to their workout view while coaches enter the web analytics portal.

Module 2: Real-Time 3D Pose Tracking

VISION ENGINE

In Simple Words: The phone camera tracks your body joints in 3D without requiring any sensors, wires, or wearable hardware.

HOW IT WORKS IN THE APP

- **30 Frames Per Second:** Runs on-device using Google ML Kit Pose Detection at full video speed.
- **33 Key Landmarks:** Tracks nose, shoulders, elbows, wrists, hips, knees, and ankles in three dimensions (X, Y, and depth Z).
- **Zero Wearables:** Everything is calculated purely from normal smartphone camera video.

Presentation Answer Tip: If asked why we chose on-device pose detection, explain that extracting 33 landmarks locally on the phone uses zero network bandwidth and runs smoothly even if internet drops.

Module 3: AI Exercise Recognition & Classification

DEEP
LEARNING

In Simple Words: The AI automatically watches you and recognizes which exercise you are doing out of 7 different workouts.

HOW IT WORKS IN THE APP

- **7 Supported Exercises:** Squat, Push-Up, Lunge, Bicep Curl, Plank, High Knees, and Jumping Jack.
- **PoseC3D 3D Convolutional Neural Network:** Connects joint points into spatiotemporal cylindrical limb heatmaps over time, capturing full movement trajectory curves.
- **Stillness Guard:** Does not guess randomly when you are standing still; only activates when you begin a genuine repetition.

Presentation Answer Tip: If asked about accuracy, state clearly that PoseC3D achieved 63.54 percent Top-1 accuracy under strictly clean, deduplicated video evaluation, with near 98 percent live recognition confidence on Squats and Push-Ups.

Module 4: Real-Time Repetition Counting

STATE MACHINE

In Simple Words: Counts your exercise repetitions automatically and rejects half-hearted, incomplete, or cheating reps.

HOW IT WORKS IN THE APP

- **4-Stage Closed State Machine:** Tracks movement through Standing, Descending, Bottom Inflection, and Ascending phases.
- **Valid vs Invalid Separation:** Only awards a full repetition point if the joint passed the required depth angle (for example, knee angle below 115 degrees on squats).
- **Plank Hold Timer:** Automatically converts to a precision isometric hold timer for planks, counting seconds held.

Presentation Answer Tip: If asked how cheating is prevented, explain that the finite state machine requires reaching the bottom inflection point before the rep counter increments. Half-reps are registered as invalid.

Module 5: Posture Correctness & Live Form Scoring

KINEMATIC
RULES

In Simple Words: Evaluates how well you perform each exercise and displays a live 0 to 100 percent form score on your screen.

HOW IT WORKS IN THE APP

- **Color-Coded Skeleton:** Shows a bright green skeleton when form is excellent, and turns yellow or red when mistakes occur.
- **Biomechanic Rule Checks:** Measures back straightness, squat depth, arm swing, elbow position, and torso lean.
- **Rep Breakdown:** Every finished repetition receives an individual score and fault summary stored in your workout log.

Presentation Answer Tip: If asked about the scoring formula, explain that every repetition begins at 100 percent and deducts points for specific kinematic deviations like torso lean or shallow depth.

Module 6: Smart Body Scanner & Measurement Tracking

ANTHROPOMETRY

In Simple Words: Measures your shoulder width, hip width, and arm span using the camera, and tracks weight transformation over time.

HOW IT WORKS IN THE APP

- **Smart Distance Guidance:** Guides the user with on-screen voice and color cues (Move closer or Step back) until body height fills the ideal framing zone.
- **Height Calibration Anchor:** Uses user entered height to convert camera pixel distances into true real-world centimeters.
- **Transformation Tracking:** Logs weight, Body Mass Index gauge, Devine ideal weight range, and exports assessment PDF reports.

Presentation Answer Tip: If asked if body measurements are fixed ratios, clarify that they are true monocular photogrammetry Euclidean measurements calibrated against user height.

Module 7: Clinical Injury Prevention Across All 7 Exercises

SPORTS
MEDICINE

In Simple Words: Detects dangerous exercise habits that cause joint damage and prevents injuries like ACL knee tears and back pain.

HOW IT WORKS IN THE APP

- **Knee Valgus (Squat & Lunge):** Tracks knee inward collapse using the clinical Munro Frontal Plane Projection Angle to protect the ACL ligament.
- **Hip Sag (Push-Up & Plank):** Detects core collapse below the McGill 162 degree line to protect the lower lumbar spine.
- **Elbow Flare (Push-Up):** Detects elbows flaring out beyond 65 degrees to prevent shoulder rotator cuff tears.
- **Torso Swing (Bicep Curl):** Flags back swinging to prevent lower spine strain.

Presentation Answer Tip: Emphasize to the panel that Module 7 covers clinical injury rules for ALL 7 exercises based on published orthopedic sports medicine standards, not just knee injuries.

Module 8: Voice Coaching Companion

AUDIO COACHING

In Simple Words: An on-device audio personal trainer that talks to you, counts your reps aloud, and shouts safety corrections instantly.

HOW IT WORKS IN THE APP

- **Real-Time Speech:** Speaks out loud: "Rep 1", "Rep 2", "Starting Squat", and "10 seconds held".
- **Priority Safety Interrupt:** If you perform a dangerous movement, it immediately cuts off rep counting and warns: "Push your knees outward!" or "Lift your hips!".
- **Zero Lag:** Runs directly through on-device native speech synthesis so you do not have to look at the screen while exercising.

Presentation Answer Tip: If asked how audio spam is avoided, explain the three-tier priority hierarchy and cooldown debouncing that stops repetitive machine-gun audio while ensuring critical safety warnings take priority.

Module 9: Coach Web Dashboard & Client Pairing

WEB
PORTAL

In Simple Words: A web portal for gym trainers to monitor multiple athletes, view workout calendars, and send feedback notes.

HOW IT WORKS IN THE APP

- **Two-Way Pairing:** Coach invites an athlete by email; athlete taps Accept on their phone to securely connect.
- **Analytics Portal:** Displays monthly attendance calendars, score progression trend charts, and granular rep-by-rep fault logs.
- **Instant Feedback & PDF:** Trainer sends notes directly to the athlete's phone notifications and downloads clinical assessment PDF reports in one click.

Presentation Answer Tip: If asked about multi-tenant privacy, explain that coaches can only view athletes who explicitly accepted their pairing invitation, keeping athlete workout records private and secure.